

26. FRUIT AND ITS COLOR USING BACKTRACKING

Aim

To write a Prolog program to identify the color of a fruit using facts, rules, and backtracking.

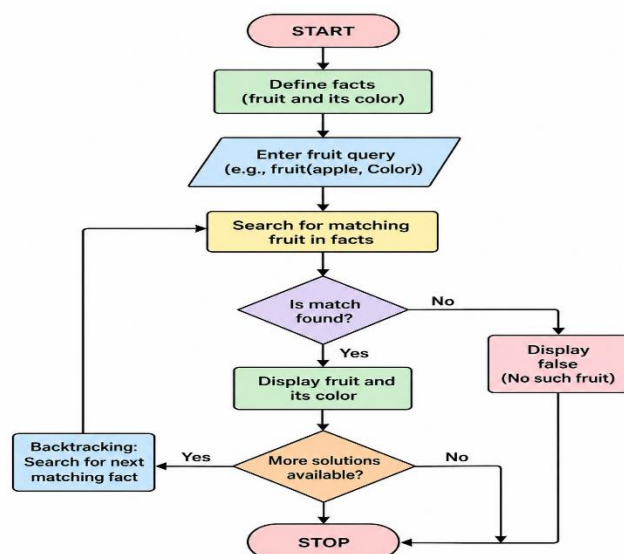
Objective

- To understand the basic concepts of Prolog.
- To represent fruits and their colors using facts.
- To use rules for querying fruit colors.
- To understand how Prolog uses backtracking to find multiple solutions.

Algorithm

1. Start the program.
2. Define facts representing fruits and their colors.
3. Define a rule fruit_color/2 to find the color of a given fruit.
4. Enter a query with a fruit name.
5. Prolog searches the facts from top to bottom.
6. If a matching fact is found, Prolog returns the corresponding color.
7. If the user requests another solution using ;, Prolog performs **backtracking** and searches for another matching fact.
8. Display the result.
9. Stop.

Flowchart



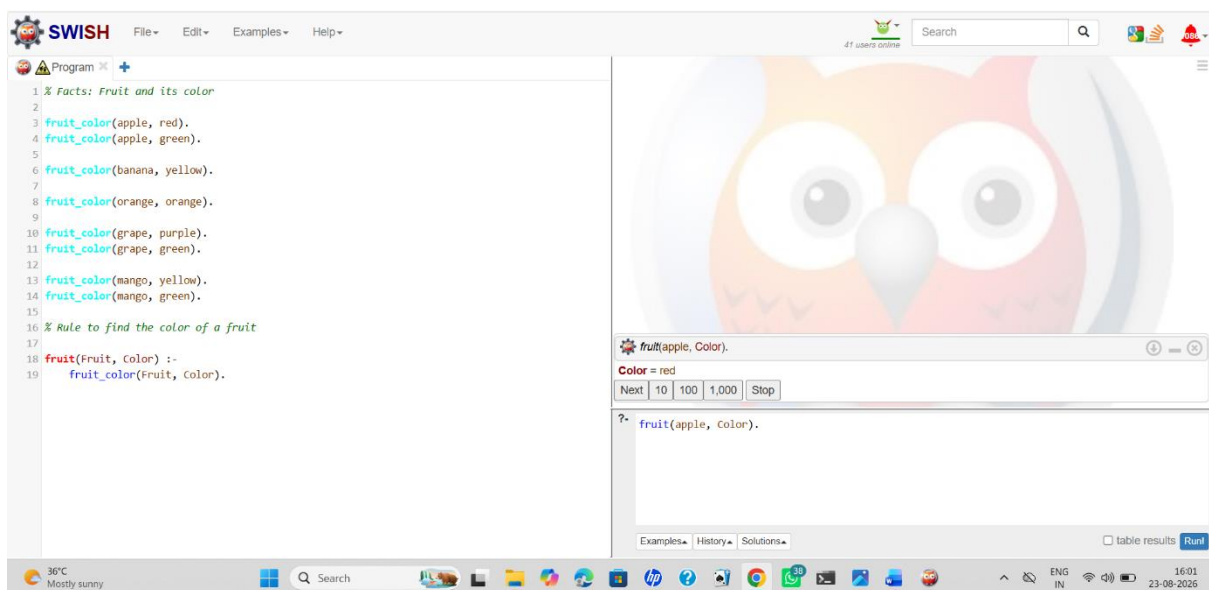
Prolog Code

```
fruit_color(apple, red).  
fruit_color(apple, green).  
fruit_color(banana, yellow).  
fruit_color(orange, orange).  
fruit_color(grape, purple).  
fruit_color(grape, green).  
fruit_color(mango, yellow).  
fruit_color(mango, green).  
fruit(Fruit, Color) :-  
    fruit_color(Fruit, Color)
```

QUERY

?- fruit(apple, Color).

.Output:



Result

Thus, the Prolog program to identify the color of a fruit using backtracking was successfully implemented and executed. Prolog returns the possible colors one by one, demonstrating the concept of backtracking.